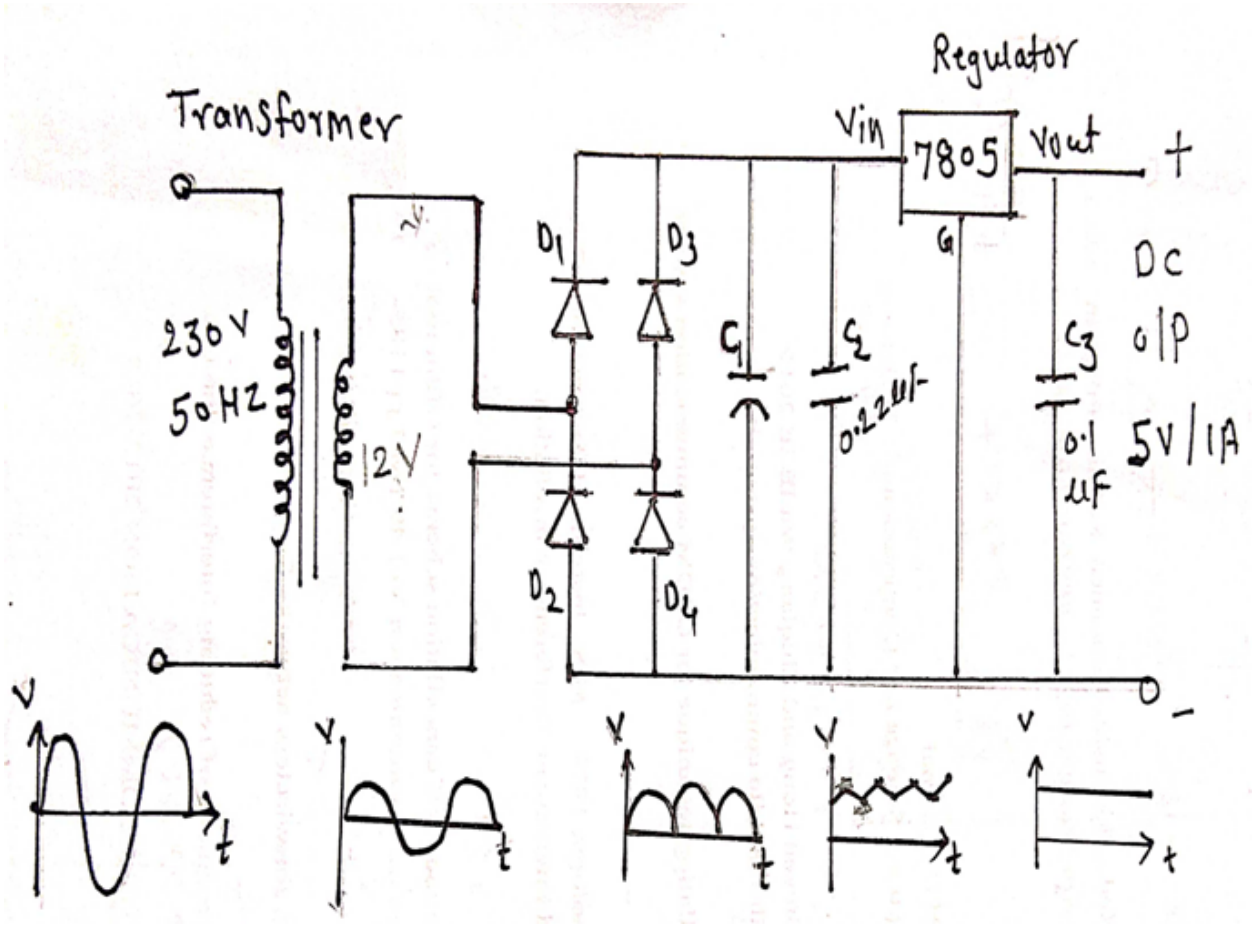
Linear Regulated Power Supply

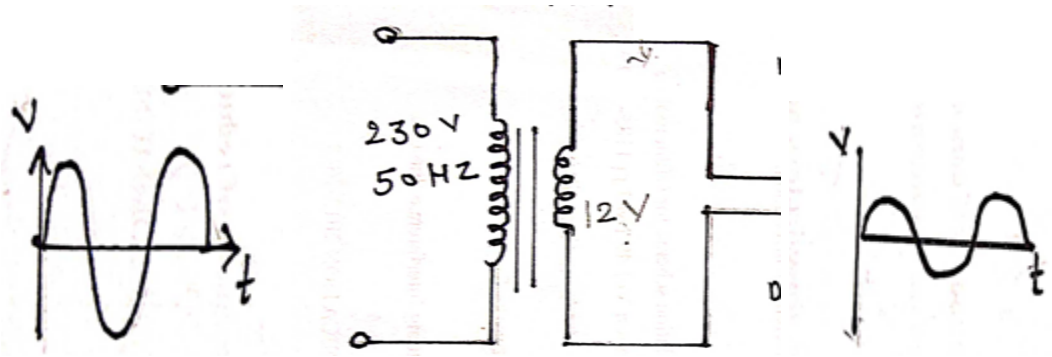
Linear regulated power supplies gain their name from the fact that they use linear, i.e. non-switching techniques to regulate the voltage output from the power supply. The term linear power supply implies that the power supply is regulated to provide the correct voltage at the output. Linear regulated power supplies also have very little ripple and very little output noise. Following fig. shows circuit diagram along with waveforms. Design of 5V/1A power supply is explained.



Working :

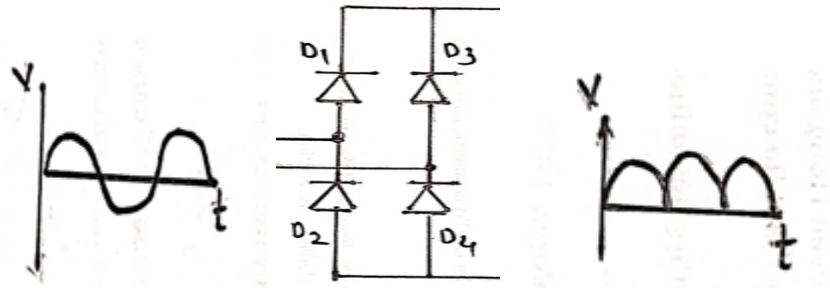
Power supply input transformer:

As many regulated power supplies take their source power from an AC mains input, it is common for linear power supplies to have a step down transformer. This also serves to isolate the power supply from the mains input for safety.



Dependent upon the rectifier approach adopted, the transformer may be a single secondary, or it may be centre tapped. Also additional windings may be present if further voltages are required. Here step down transformer of rating 230V/50Hz primary and 12V/2A secondary winding is used.

Rectifier: For converting AC into pulsating DC rectifier circuit is used. Here bridge rectifier is used**.** Diodes D1,D2 D3,D4 forms bridge rectifier



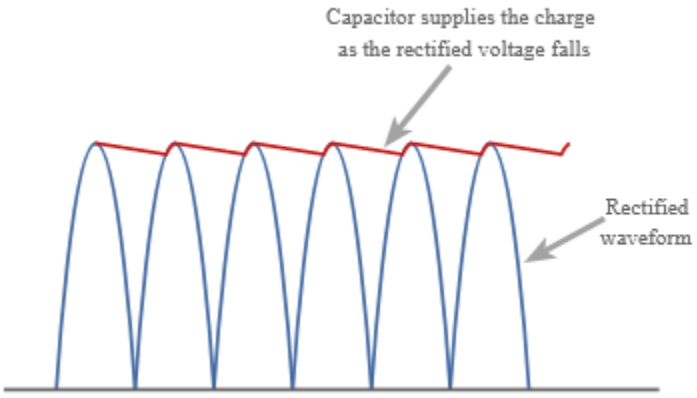
Diodes D1 and D4 conducts in positive half cycle and D2 ,D3 conducts in negative half cycle. Thus it provides output in both cycles.

Selection criterion for diodes:

For selecting diode we consider forward voltage drop, forward current and PIV (Peak Inverse voltage) rating .

for 5V/1A , diode selected is 1N4007 whose forward current is 1.5 A and PIV =1000v

Filter Capacitor: Once rectified from an AC signal, the DC needs to be smoothed to remove the varying voltage level. Large electrolytic capacitors are used for this.



The smoothing element of the circuit uses a large capacitor. This charges up as the incoming waveform from the rectifier rises to its peak. As the voltage of the rectified waveform falls away, once the voltage is below that of the capacitor, the capacitor starts to supply charge, holding the voltage up, until the next rising waveform from the rectifier.

The smoothing is not perfect and some residual ripple is present which is then removed by regulator IC

Capacitive Filter calculation

Vrms =12 V (secondary voltage)

12-1.2 = 10.8V (considering -10% variation)

10.8 *1.414 = 15.27 (DC votage)

Considering two diodes drop, peak voltage is given by

Vp= 15.27-1.4

= 13.87 V

we have ,

C=I*t /V

C= 1*10 / 6.87 = 1455.60 uF

= 2182.6 uF(considering 50 % higher)

= 2200 uF(Standard value)

where I is load current(1A) and V is difference between maximum and minimum voltage of capacitor i. e. (13.87- 7)

Voltage across capacitor = 13.87 V

= 16.64V (considering 20% higher)

capacitor selected is 2200uF/ 25 V

Regulator: Selection of regulator IC is based on following parameters

1) Output voltage 2) Output Current 3)dropout voltage 4)ripple rejection ratio

We want +5V output and current is 1 A. So we select **IC7805** which gives constant positive 5 V output and current upto 1.5A. Its dropout voltage is 2V and Ripple rejection ratio is 73 dB.

Advantages of LRPS:

Ease of use

free of any switching noise

having ripple rejection capability

low voltage noise

Disadvantages:

Low efficiency if input output voltage drop is large

Low efficiency = significant heat dissipation . May require heat sink

capable exclusively of step down operations

Size is more

Applications:

- low noise amplifiers.
- signal processing.
- data acquisition - including sensors, multiplexers, A/D converters, and sample & hold circuits.
- automatic test equipment.
- laboratory test equipment.
- control circuits.